

Minor Project 2

Customer Segmentation using K-Means Clustering

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Course: Master of Computer Applications (MCA),

Section: B

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1. Title

Customer Segmentation using K-Means Clustering (Unsupervised Machine Learning)

2. Problem Statement

Retail businesses often struggle to understand customer purchasing behavior because customer data is usually unlabeled. Without proper customer segmentation, it becomes difficult to identify different customer groups and design targeted marketing strategies.

The objective of this project is to segment mall customers into distinct groups based on their Annual Income and Spending Score using an unsupervised machine learning approach (K-Means Clustering), since no predefined labels are available for these customers. These segments can help a business design personalized marketing campaigns and improve customer retention.

3. Dataset Description

The dataset used is the Mall Customer Segmentation Dataset (Kaggle), containing 1000 customer records with the following attributes: CustomerID, Gender, Age, Annual Income (k\$), and Spending Score (1-100). It contains no predefined labels, making it suitable for unsupervised clustering.

Column	Description
CustomerID	Unique identifier for each customer
Gender	Male / Female
Age	Age of the customer (in years)
Annual Income (k\$)	Annual income of the customer, in thousand dollars
Spending Score (1-100)	Score assigned based on customer spending behavior

4. Data Preprocessing Steps

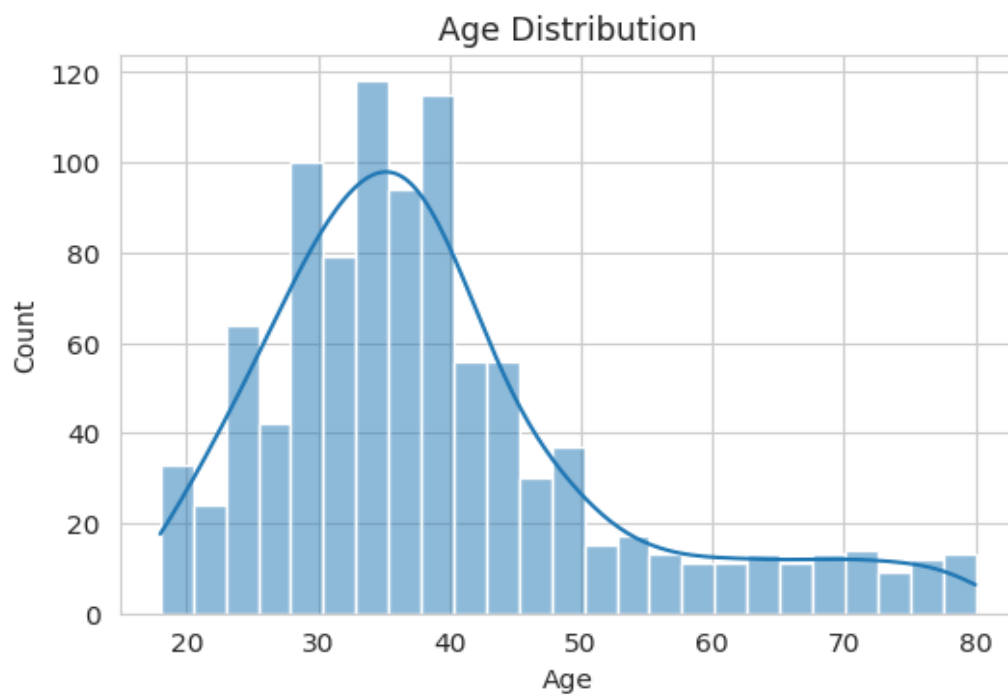
Before clustering, the dataset was cleaned using the following steps:

- **Missing Value Treatment:** Numeric columns (Age, Annual Income, Spending Score) had missing values filled using the median, since median is robust to outliers and skewed data.
- **Categorical Handling:** Missing values in Gender were filled using the mode (most frequent value).
- **Duplicate Check:** The dataset was checked for duplicate records; none were found.
- **Feature Scaling:** Numeric features were standardized using StandardScaler (zero mean, unit variance), since K-Means relies on Euclidean distance and is sensitive to differences in feature scale.

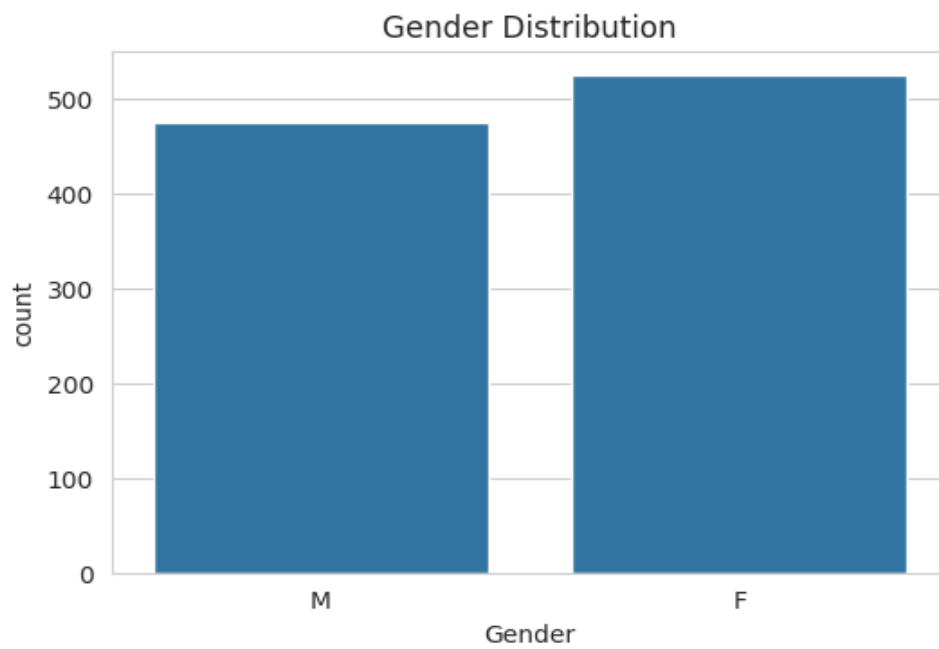
5. Exploratory Data Analysis

Visualizations were created to study feature distributions, detect outliers, and identify natural groupings among customers before clustering.

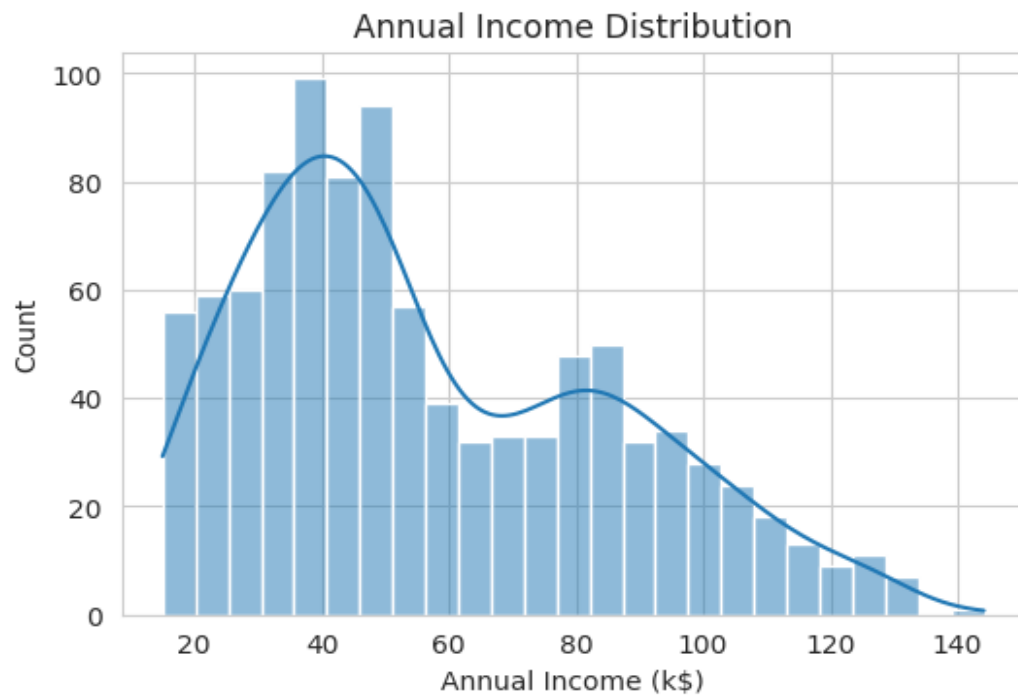
5.1 Age Distribution



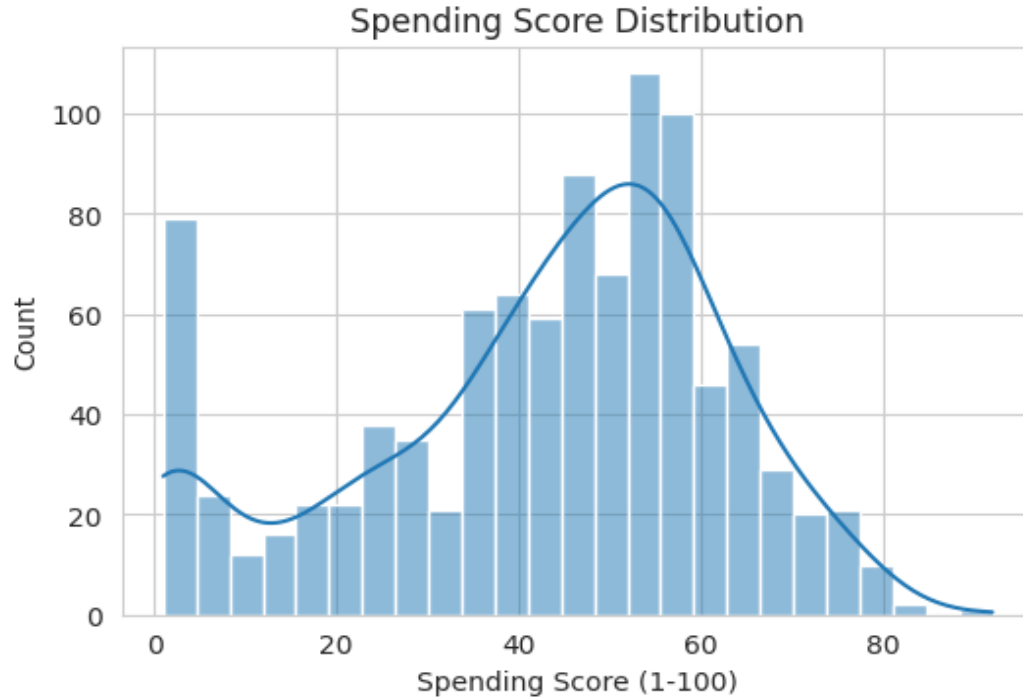
5.2 Gender Distribution



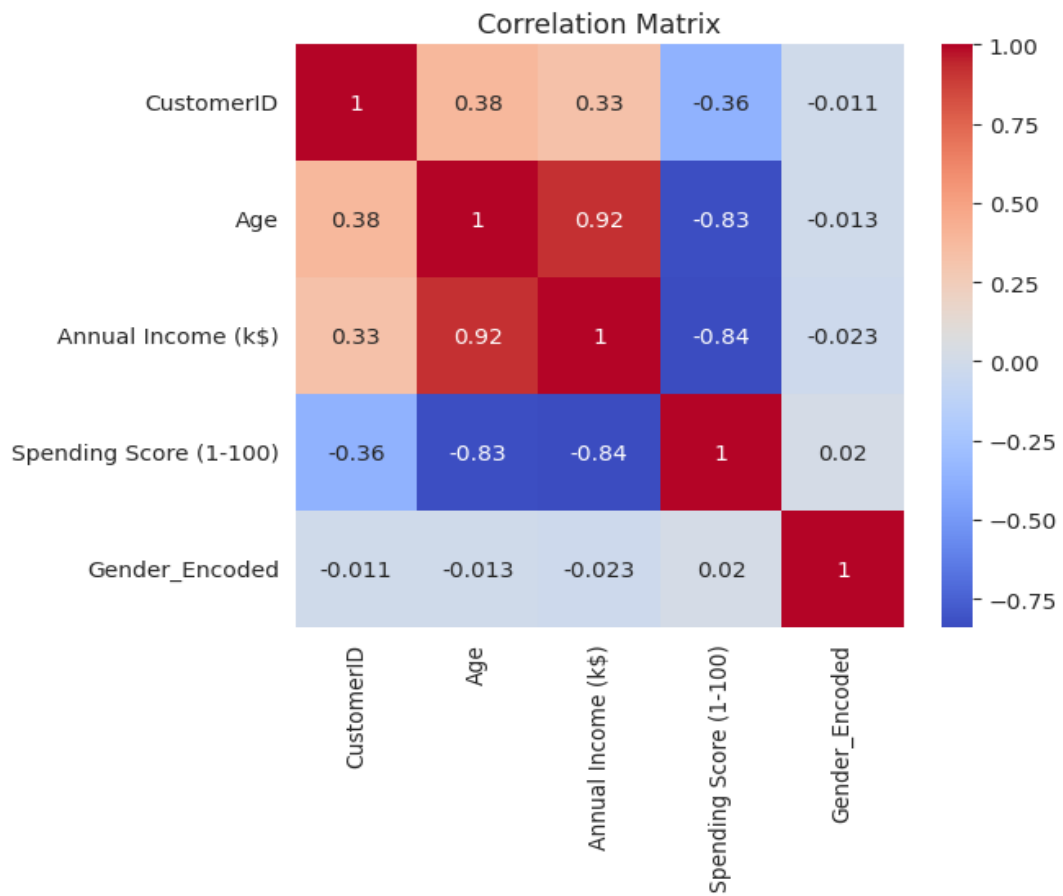
5.3 Annual Income Distribution



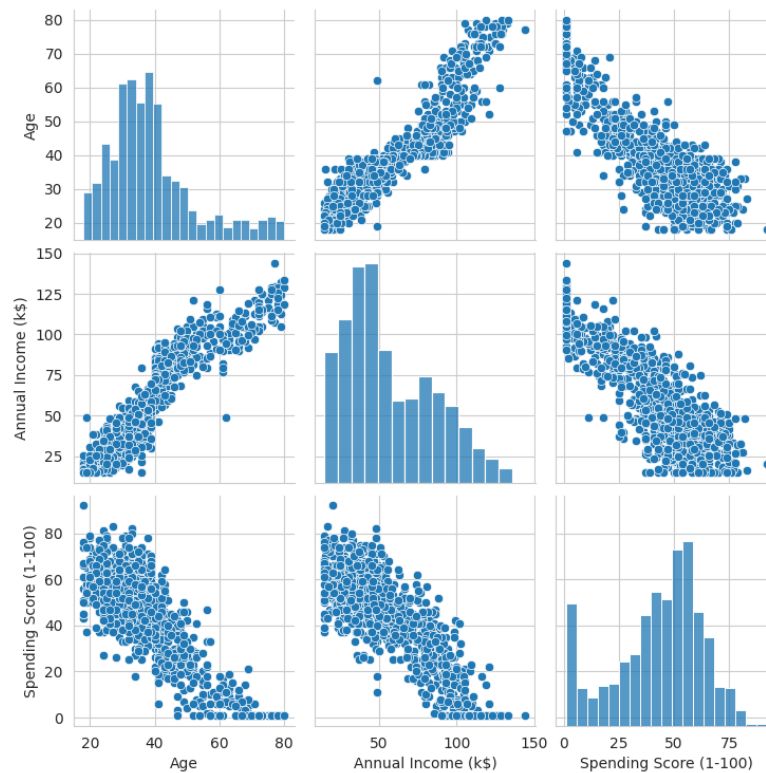
5.4 Spending Score Distribution



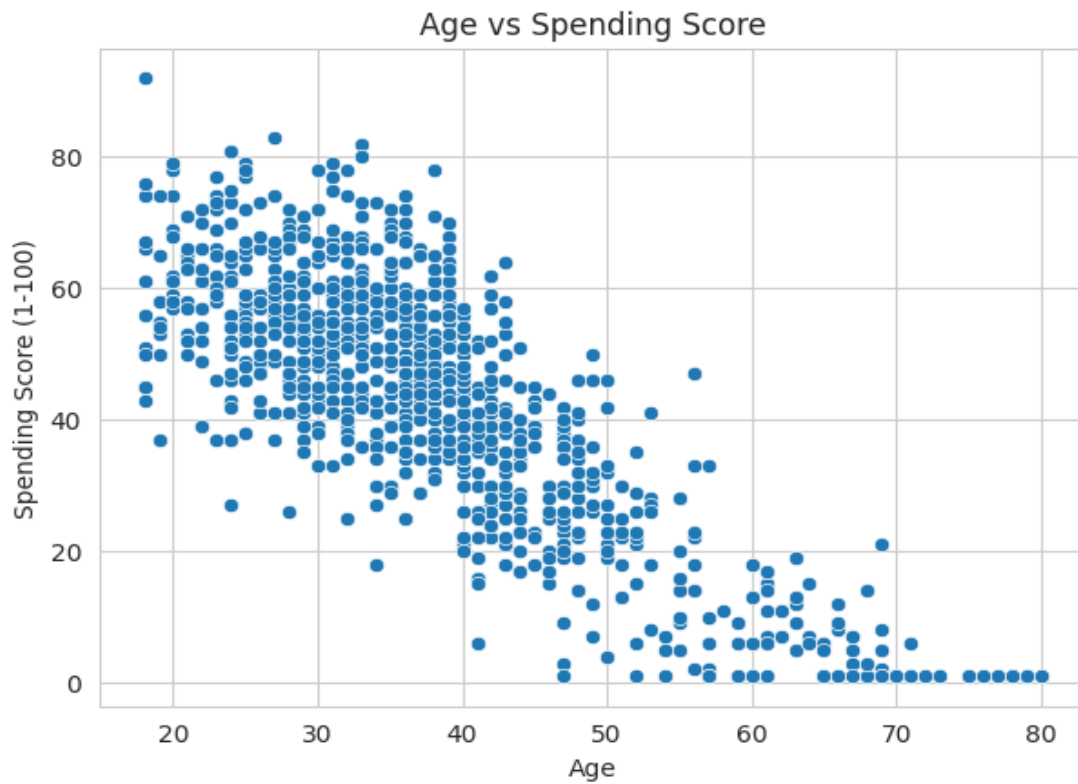
5.5 Correlation Heatmap



5.6 Pairwise Relationships



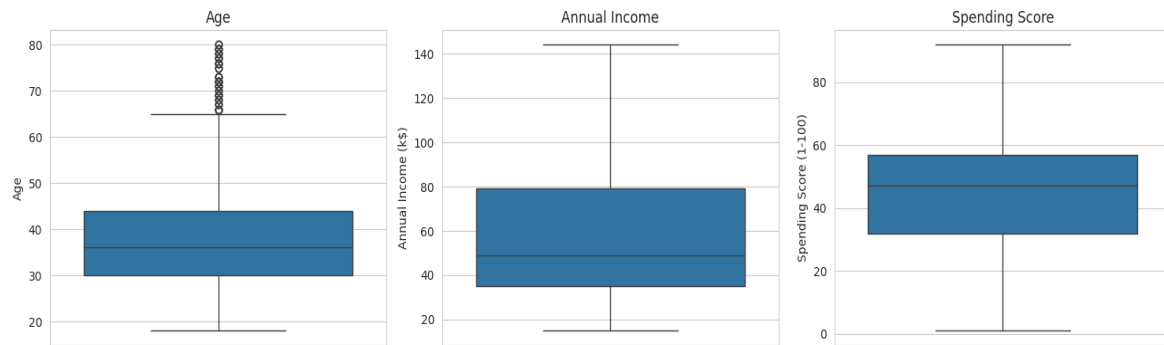
5.7 Age vs Spending Score



5.8 Annual Income vs Spending Score



5.9 Outlier Check (Boxplots)



5.10 Key Observations

- The dataset contains customers of different age groups.
- Annual income is distributed across low, medium, and high-income customers.
- Spending scores vary significantly, indicating different purchasing behaviors.
- No strong linear correlation exists among the numerical features.
- The scatterplot of Annual Income vs Spending Score shows customers forming visually separable groups, confirming that K-Means clustering is a suitable approach for this data.

6. Model Used

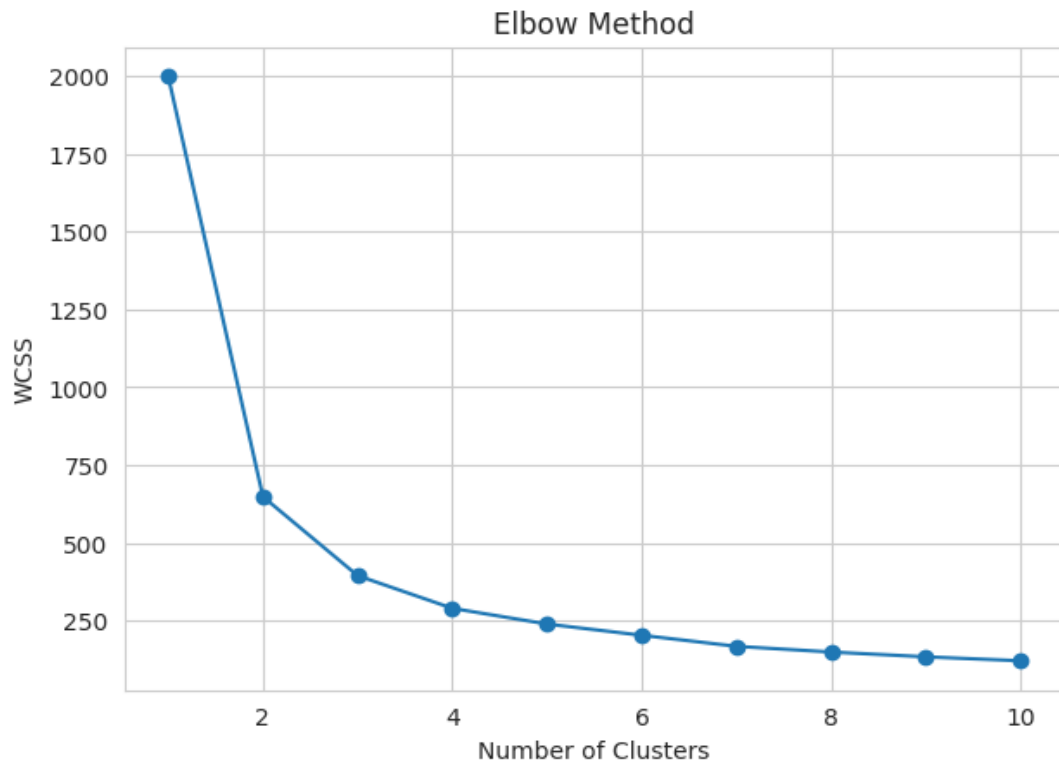
K-Means Clustering was chosen for this project because:

- It performs well on numeric, continuous features like Income and Spending Score.
- It works best for compact, roughly spherical clusters, which matches the pattern seen in the Income vs Spending Score scatter plot.
- Annual Income and Spending Score were selected as the clustering features since they are the most business-relevant for segmenting customers by purchasing behavior.

7. Training Process

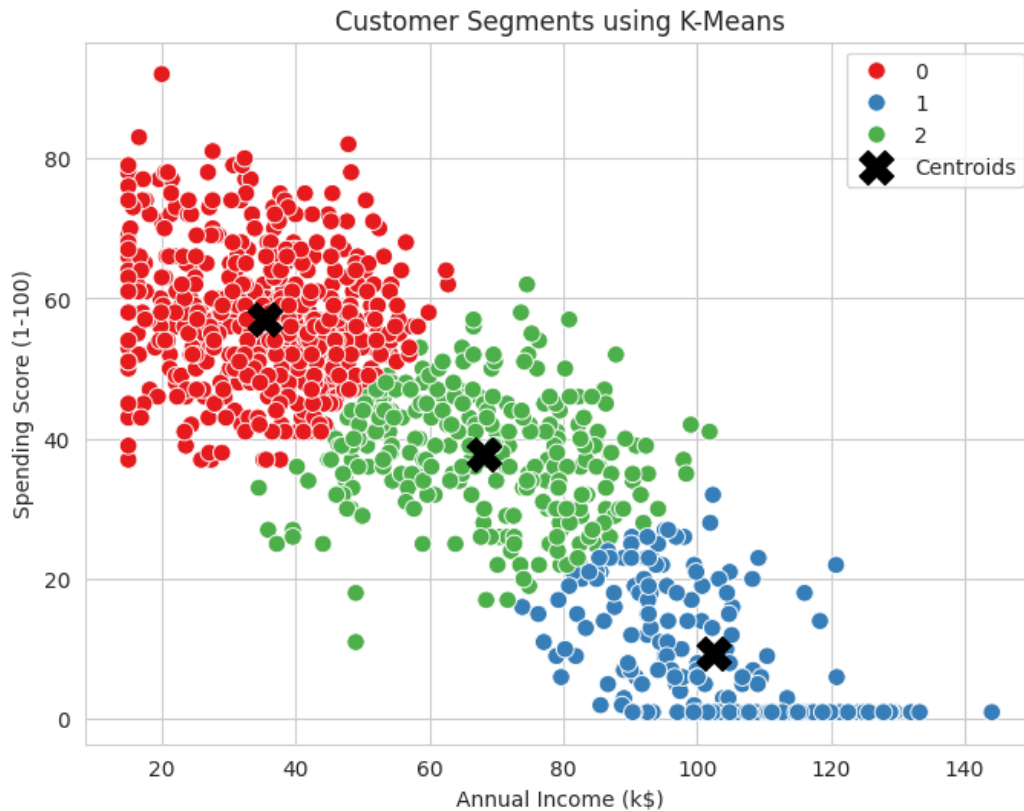
- The Elbow Method was used to find the optimal K by training K-Means for K = 1 to 10 and recording the WCSS (Within-Cluster Sum of Squares).
- Silhouette Scores were also computed for each K to confirm the best choice.
- The final model was trained with `n_clusters=3`, `init="k-means++"`, and `random_state=42`.
- Cluster labels were assigned to each customer, and centroids were computed.

7.1 Elbow Method



The Elbow Method shows a sharp decrease in WCSS from $K = 1$ to $K = 3$. After $K = 3$, the curve starts to flatten, indicating diminishing returns from adding more clusters. This is also supported by the Silhouette Scores, where $K = 3$ gives one of the highest and most stable scores among practical cluster counts. Therefore, the optimal number of clusters selected for this project is $K = 3$.

7.2 Final Clusters with Centroids



8. Evaluation Metrics and Results

Since there are no ground-truth labels, classification metrics (accuracy, precision, recall, F1-score) do not apply. Clustering-specific metrics were used instead:

- **Silhouette Score (0.474):** Measures how well-separated and cohesive the clusters are, ranging from -1 to 1. A score of 0.474 indicates reasonably well-defined, moderately separated clusters.
- **Davies-Bouldin Index (0.720):** Measures average similarity between each cluster and its most similar cluster (lower is better). A value below 1 suggests good cluster quality.

8.1 Cluster Sizes

Cluster	Number of Customers
0	527
1	183
2	290

8.2 Cluster Profile Summary (Average Values)

Cluster	Avg. Age	Avg. Annual Income (k\$)	Avg. Spending Score
0	30.1	35.3	57.0
1	61.1	102.5	9.3
2	40.8	68.1	37.7

8.3 Cluster Interpretation

Based on the cluster centroids and average feature values, the three customer segments are interpreted as follows:

- **Cluster 0 - Young, Moderate Income, High Spenders:** Average age around 30, moderate income (~\$35k), but high spending score (~57). These are likely younger, trend-driven customers who spend actively despite moderate income - good targets for loyalty programs and trend-based promotions.
- **Cluster 1 - Older, High Income, Low Spenders:** Average age around 61, high income (~\$102k), but very low spending score (~9). These customers earn well but spend very little in the mall - potential targets for re-engagement or premium/exclusive offers to encourage more spending.
- **Cluster 2 - Middle-Aged, Moderate-High Income, Average Spenders:** Average age around 41, moderate-high income (~\$68k), and average spending score (~38). This is the largest and most typical customer segment, suitable for general marketing campaigns.

These insights allow the business to design differentiated marketing strategies instead of a one-size-fits-all approach.

9. Conclusion

The K-Means clustering algorithm successfully segmented customers into three meaningful groups based on Annual Income and Spending Score.

The Elbow Method, supported by Silhouette Scores across different values of K, indicated that three clusters provide the best balance between simplicity, compactness, and separation for this dataset.

The Silhouette Score (0.474) and Davies-Bouldin Index (0.720) confirm that the clusters are reasonably well-defined and well-separated.

Each cluster represents a distinct customer segment with different income and spending patterns, which can help the business design targeted marketing strategies - such as loyalty programs for high spenders, re-engagement campaigns for high-income low spenders, and general campaigns for the average customer segment.

10. References

- Kaggle - Mall Customer Segmentation Dataset -[link](#)
- Scikit-learn Documentation - <https://scikit-learn.org/stable/modules/clustering.html#k-means>
- Pandas Documentation - <https://pandas.pydata.org/>
- Matplotlib Documentation - <https://matplotlib.org/>
- Seaborn Documentation - <https://seaborn.pydata.org/>

Theory Questions

Q1. Semantic vs Instance vs Panoptic Segmentation

- **Semantic Segmentation:** Assigns a class label to every pixel in an image, but does not distinguish between different objects of the same class. For example, all cars in an image are labeled as "car" without separating individual cars.
- **Instance Segmentation:** Detects and segments each individual object separately, even if they belong to the same class. For example, each car in the image gets its own separate mask/label (Car 1, Car 2, etc.), but background regions like sky or road are not labeled.
- **Panoptic Segmentation:** Combines both semantic and instance segmentation. It labels every pixel with a class (like semantic segmentation) while also separating individual object instances (like instance segmentation). This means both countable objects (cars, people) and uncountable background regions (sky, road) are labeled in a single unified output.

Q2. What is Fuzzy Logic, and how is it different from Boolean Logic?

Fuzzy Logic is a form of reasoning that deals with degrees of truth rather than fixed true/false values. Instead of a value being strictly 0 or 1, fuzzy logic allows values to range anywhere between 0 and 1, representing partial truth or degree of membership in a set.

Difference from Boolean Logic:

- **Boolean Logic:** Works with only two crisp values - True (1) or False (0). There is no in-between state. For example, "Is the water hot?" can only be answered Yes or No.
- **Fuzzy Logic:** Allows partial truth values between 0 and 1. For example, water at 60 degrees could be "70% hot" and "30% warm" at the same time, which better reflects real-world uncertainty and gradual transitions.

Fuzzy logic is widely used in systems that require human-like decision-making under uncertainty, such as air conditioners, washing machines, and control systems, whereas Boolean logic is used in traditional digital circuits and binary decision-making.

Q3. Customer Dataset with Age, Annual Income, and Spending Score (No Labels)

1. Is this a supervised or unsupervised learning problem?

This is an unsupervised learning problem. Since there are no predefined labels or target categories for the customers, the model must discover natural groupings or patterns in the data on its own, without any prior guidance about the correct output.

2. Which clustering algorithm would you choose first and why?

K-Means Clustering would be the first choice. It is simple, computationally efficient, and works well on numeric, continuous data such as Age, Annual Income, and Spending Score. It is also easy to interpret and visualize, and the optimal number of clusters can be determined using the Elbow Method or Silhouette Score, making it a practical first approach before trying more complex algorithms like DBSCAN or Hierarchical Clustering.

3. What assumptions does that algorithm make about the data?

- Clusters are assumed to be roughly spherical (convex) and similarly sized in feature space.
- Features are assumed to be equally important and are best used after scaling, since K-Means relies on Euclidean distance.
- The number of clusters (K) must be specified in advance.
- Clusters are assumed to have similar density and variance.
- The algorithm is sensitive to outliers, since extreme values can distort centroid positions.